How-To Library

Informatica

Ingesting SQL Server data and CDC changes into Amazon Redshift with Mass Ingestion Databases

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Abstract

This article shows how to set up a Mass Ingestion Databases job that performs an initial load of SQL Server data to an Amazon Redshift target and then replicates the incremental CDC changes to the target.

Supported Versions

- Informatica Intelligent Cloud Services November 2023

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Overview

This article walks through the steps required to initially set up a Mass Ingestion Databases environment and then configure and implement a database ingestion combined initial and incremental load job that replicates SQL Server data to an Amazon Redshift target. The initial load portion of the job initially populates the Amazon Redshift target with point-in-time data from SQL Server. The incremental load portion of the job replicates the SQL Server change data in real time to keep the source and target in sync.

The use cases for replicating real-time change data from a mission-critical SQL Server production system to an Amazon Redshift target include:

- Minimizing intrusive, non-critical work, such as offline reporting or analytical operations on the SQL Server source database system.
- Reducing costly CPU usage, disk space, and CDC processing time on the SQL Server source database system.
- Getting the latest database changes to your data warehouse in real time for downstream processing. After an initial batch load of data to the data warehouse, Mass Ingestion Databases can propagate data changes continuously from a source database to keep the data in the data warehouse up to date.
- Migrating to a subscription-based Cloud environment.

For more information, see the Cloud Mass Ingestion documentation on the Doc Portal at https://docs.informatica.com/integration-cloud/cloud-mass-ingestion/current-version.html.

Task flow

Perform the following tasks, in the order shown, to implement SQL Server to Amazon Redshift combined initial and incremental load processing.

1. "Initial setup of your environment" on page 3.
2. "Creating projects and project folders" on page 10 in which to store your database ingestion tasks.
3. Prepare and learn about "SQL Server sources" on page 10 and "Amazon Redshift targets" on page 13.
4. "Configuring connections" on page 13 if not already available.
5. "Configuring a database ingestion task" on page 19.
6. "Deploying the database ingestion task" on page 29.
7. "Running the deployed database ingestion job" on page 29.
8. "Monitoring database ingestion jobs" on page 29.

Initial setup of your environment

You must complete some initial setup tasks in your Informatica Intelligent Cloud Services environment before you start configuring connections and database ingestion tasks.

1. Verify that your source and target versions are supported.
2. Check that you system meets the minimum system requirements.
3. Create an organization and add users to it.
4. Download and install a Secure Agent.
5. Configure the Database Ingestion agent service for your runtime environment.

Step 1: Verify that your source and target versions are supported

To determine the source and target types and versions that are supported for each ingestion type, check the KB article What are the supported sources and targets for IICS Cloud Mass Ingestion service?.

To determine the web browsers that are compatible with Mass Ingestion and the operating systems that are supported for the Secure Agent, check the Product Availability Matrix (PAM) for Informatica Intelligent Cloud Services (IICS) on the Informatica Network at https://network.informatica.com/community/informatica-network/product-availability-matrices/.

Step 2: Check Mass Ingestion Databases system requirements

The following table lists the Mass Ingestion Databases minimum system requirements for the Secure Agent:

Component	Minimum requirement
Cores per CPU	8 minimum, 16 recommended if you need to process a large number of source tables in an initial load
Memory	32 GB
Disk space	5 GB per job, based on a row size of 2 KB

Step 3: Set up an organization and organization users

An organization is a secure area within the Informatica Intelligent Cloud Services repository that stores your licenses, user accounts, ingestion tasks, and information about jobs and security. Set up an organization in Administrator.

If your organization has the Organization Hierarchy license, you can also create one or more sub-organizations within your organization. You can create sub-organizations to represent different business environments in your company. For example, you might create separate sub-organizations to represent your development, testing, and production environments.

Perform the following tasks:

- On the **Organization** page, configure organization properties such as the organization name and address, authentication information, and notification email addresses. Click the Sub-organizations tab to add a sub-organization to the organization.
- On the **Users** and **User Groups** pages, add and configure users and user groups for the organization. On the User Roles page, you can configure user role permissions, which can be assigned to users and user groups.

Administrator provides context-sensitive help for all of these pages. To display the help for the page in which you're working, click the help (?) icon and select **Online Help**.

Step 4: Download and install a Secure Agent

A Secure Agent is a lightweight program that runs tasks and enables secure communication across the firewall between your organization and Informatica Intelligent Cloud Services.

Install the Secure Agent on all the systems from which you want to use Mass Ingestion Databases. You can install a Secure Agent on Windows or Linux.

When you download and install a Secure Agent, a *Secure Agent group*, also called a runtime environment, is created. A Secure Agent group can contain one Secure Agent or multiple agents.

Note: Mass Ingestion Databases does not support the Hosted Agent or serverless runtime environment.

Downloading and installing the Secure Agent on Linux

To install the Secure Agent on a Linux machine, download and run the Secure Agent installer and then register the agent.

Before you download and install the Secure Agent, verify that no other Secure Agent is installed on the machine using the same Linux user account. If any other Secure Agent exists, you must uninstall it first.

Tip: To verify the checksum of the Secure Agent installation program, use the agent REST API version 2 resource. For more information about the agent resource, see *REST API Reference*.

1. Open Administrator and select **Runtime Environments**.

2. On the **Runtime Environments** page, click **Download Runtime Installer**.
3. Select **Secure Agent** as the Environment Type.
4. Select **Linux 64** as the Platform.
5. Click **Copy** to copy the install token.

   You will need this token to install the Secure Agent. The token expires after 24 hours. If the token expires before you install the Secure Agent, repeat steps 1 to 5 to obtain a new token. You don't need to download the installer again.
6. Click **Download**.

   The installer is downloaded to your machine. The name of the installer is `agent64_install_ng_ext.<agent core version>.bin`.
7. Save the installer to a directory on the machine where you want to run the Secure Agent.

   Note: Ensure that the file path doesn't contain spaces or multibyte characters. If the file path contains spaces, the installation might fail. If the path contains multibyte characters, the Secure Agent might not start.
8. From a shell command line, navigate to the directory where you downloaded the installation program and enter the following command:

   ```
   ./agent64_install_ng_ext.bin -i console
   ```
9. When the installer completes, navigate to the following directory:

   ```
   <Secure Agent installation directory>/apps/agentcore
   ```
10. To start the Secure Agent, enter the following command:

    ```
    ./infaagent.sh startup
    ```

    The Secure Agent Manager starts. Register the agent using the user name that you use to access Informatica Intelligent Cloud Services and also supply the install token.
11. To register the agent, in the `<Secure Agent installation directory>/apps/agentcore` directory, enter one of the following commands using your Informatica Intelligent Cloud Services user name and the token that you copied:
    - To add the agent to its own Secure Agent group, use the following command:

      ```
      ./consoleAgentManager.sh configureToken <user name> <install token>
      ```
    - To add the agent to an existing Secure Agent group, use the following command:

      ```
      ./consoleAgentManager.sh configureTokenWithRuntime <user name> <install token>
      <Secure Agent group name>
      ```

      Note: If the command includes a Secure Agent group name that doesn't exist, the Secure Agent is not assigned to a group. Be sure to use a valid Secure Agent group name.

    The following table lists the command options:

Option	Description
User Name	Required. Informatica Intelligent Cloud Services user name of the user installing the Secure Agent.
Install Token	Required. The install token that you copied when you downloaded the runtime installer.
Secure Agent group name	Optional. Include a group name when you want to add the agent to an existing Secure Agent group instead. If this option isn't included in the command, the agent will be in its own Secure Agent group.

You can check the registration status of a Secure Agent using the following command:

```
./consoleAgentManager.sh isConfigured
```

Downloading and installing the Secure Agent on Windows

To install the Secure Agent on a Windows machine, you must download and run the Secure Agent installation program and then register the agent.

Secure Agent registration requires an install token. To get the install token, copy the token when you download the agent or use the **Generate Install Token** option in Administrator. The token expires after 24 hours.

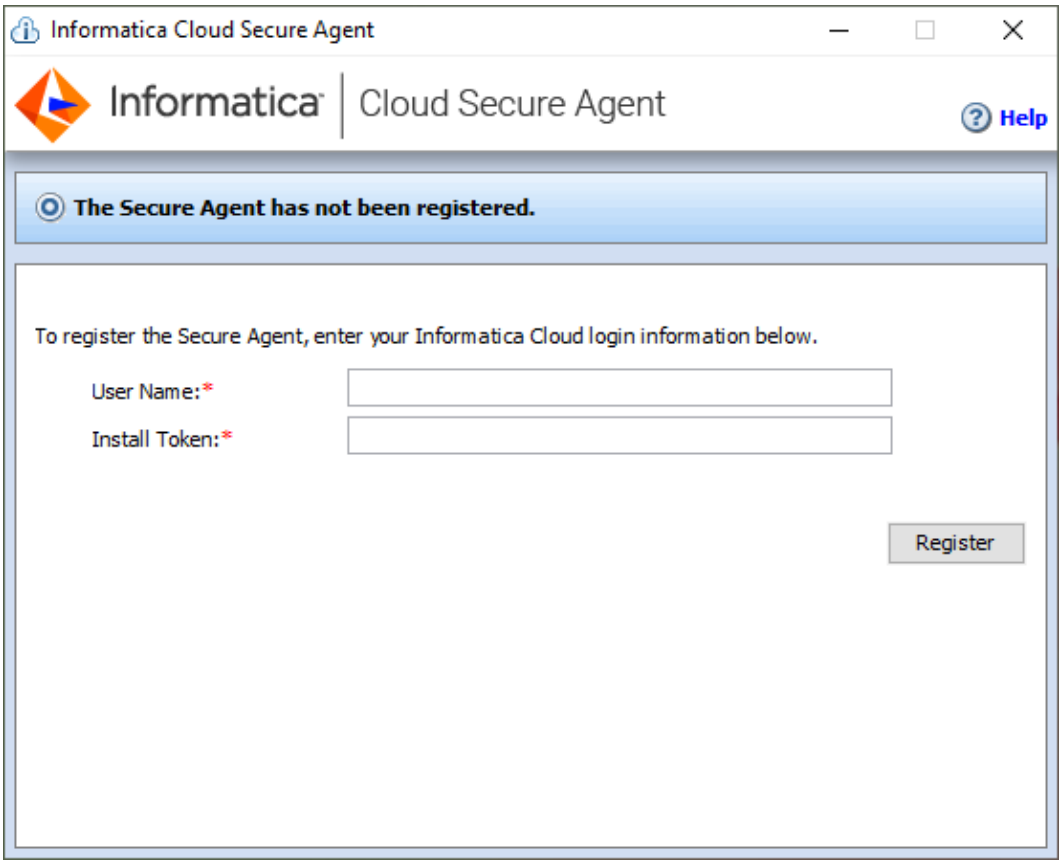
Before you download and install the Secure Agent, verify that no other Secure Agent is installed on the machine. If there is, you must uninstall it.

Tip: To verify the checksum of the Secure Agent installation program, use the agent REST API version 2 resource. For more information about the agent resource, see *REST API Reference*.

1. Open Administrator and select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Download Secure Agent**.
3. Select the Windows 64-bit operating system platform, copy the install token, and then click **Download**.

   The installation program is downloaded to your machine. The name of the installation program is `agent64_install_ng_ext.<agent core version>.exe`.
4. Run the installation program as an Administrator:
   a. Specify the Secure Agent installation directory, and click **Next**.
   b. Click **Install** to install the agent.

The Secure Agent Manager opens and prompts you to register the agent as shown in the following image:



5. If you did not copy the install token when you downloaded the agent, click **Generate Install Token** on the **Runtime Environments** page in Administrator, and copy the token.
6. In the Secure Agent Manager, enter the following information, and then click **Register**:

Option	Description
User Name	User name that you use to access Informatica Intelligent Cloud Services.
Install Token	Token that you copied.

The Secure Agent Manager displays the status of the Secure Agent. It takes a minute for all of the services to start.

7. If your organization uses an outgoing proxy server to connect to the internet, enter the proxy server information.
8. Close the Secure Agent Manager.

The Secure Agent Manager minimizes to the taskbar and continues to run as a service until stopped.

Step 5: Configure the Database Ingestion service for your runtime environment

Mass Ingestion Databases uses the Database Ingestion agent service to run database ingestion jobs. The Database Ingestion service is automatically downloaded to the system when you download the Secure Agent.

After you download the Secure Agent to your runtime environment and enable the Database Ingestion service, the Database Ingestion packages are pushed to the on-premises system where the Secure Agent runs. You can then optionally configure properties for the Database Ingestion service that runs on the Secure Agent.

Database Ingestion service properties

To change or optimize the behavior of the Database Ingestion service that your Secure Agent group uses, configure Database Ingestion properties for your runtime environment.

To configure the properties, open your runtime environment and click **Edit**. Under **System Configuration Details**, select the **Database Ingestion** service and the **DBMI_AGENT_CONFIG** type.

The following table describes the Database Ingestion agent service properties:

Property	Description
maxTaskUnits	The maximum number of database ingestion task units that can run concurrently on an on-premises machine where the Secure Agent is running. Task units are not related to the capacity and availability of your hardware or software. You can configure maxTaskUnits to precisely control CPU usage. Valid values are 1 to 2000000000 (2 billion). To calculate a reasonable number of task units for your Secure Agent machine, Informatica recommends that you divide the number of cores by 3 or 4. For example, if you have an 8-core machine, you could set this property to 2. Then monitor CPU usage and adjust the property value as needed to tune performance. During initial load processing, this property determines the number of tables that can be unloaded simultaneously. Remaining tables are queued and start unload processing when resources become available. **Note:** A single job can process many tables. The total number of tables that can be processed is limited only by available memory. On the average, 25 MB of RAM is required per table for an initial load task based on a 1 KB row size. During incremental load processing, this property determines the number of database ingestion jobs that can run simultaneously. Setting this property to a value greater than the number of cores on the Secure Agent machine can increase parallelism for task execution but also cause performance bottlenecks at task execution time.
serviceLogRetentionPeriod	The number of days to retain each internal Database Ingestion service log file after the last update is written to the file. When this retention period elapses, the log file is deleted. The default value is 7 days. Service logs are retained on the Secure Agent host where they are created: `<infaagent>/apps/Database_Ingestion/logs`.

Property	Description
taskLogRetentionPeriod	The number of days to retain each job log file after the last update is written to the file. When this retention period elapses, the log file is deleted. The default value is 7 days.
serviceUrl	The URL that the Database Ingestion service uses to connect to the Informatica Intelligent Cloud Services cloud.
logLevel	The level of detail to include in the logs that the Database Ingestion service produces. Options are: - TRACE - DEBUG - INFO - WARN - ERROR The default value is TRACE.
taskExecutionHeapSize	The maximum heap size, in gigabytes, for the Task Execution service. This value, in conjunction with maxTaskUnits property, affects the number of concurrent database ingestion tasks that can run on a Secure Agent. Try increasing the heap size to run more tasks concurrently. Enter this value followed by "g" for gigabytes, for example, '9g'. The default value is '8g'.
useProxy	Set this property to true to enable the DBMI Agent to go through a proxy when connecting to or writing data to targets. The DBMI Agent then uses the proxy settings from the Secure Agent proxy configuration. By default, proxy settings are not used.
intermediateStorageDirectory	For incremental load and combined initial and incremental load jobs, the local root directory under which intermediate files that contain data are stored when the **Enable Persistent Storage** option is selected in the associated task definitions.
storageBackupDirectory	For incremental load and combined initial and incremental load jobs, the path to the directory that stores backup files when the **Enable Persistent Storage** option is selected in the associated task definitions.
storageProperties	For incremental load and combined initial and incremental load jobs, a comma-separated list of key=value pairs that is used when the **Enable Persistent Storage** option is selected in the associated task definitions. Specify this property only at the direction of Informatica Global Customer Support.
task_container.jvm.allowExceptionForInvalidEncodedData	If you receive transliteration errors that report invalid encoding to UTF-8, and you do not want to repair or correct the source data, set this property to false so that database ingestion jobs do not fail when trying to unload the data from the source. With this setting, the Database Ingestion service passes an equivalent Java property to the DataDirect JDBC driver to prevent the exception from occurring. After you set this property, you must restart the Database Ingestion service.

Database Ingestion service environment variable

To change or optimize the behavior of the Database Ingestion Agent, define the following environment variable:

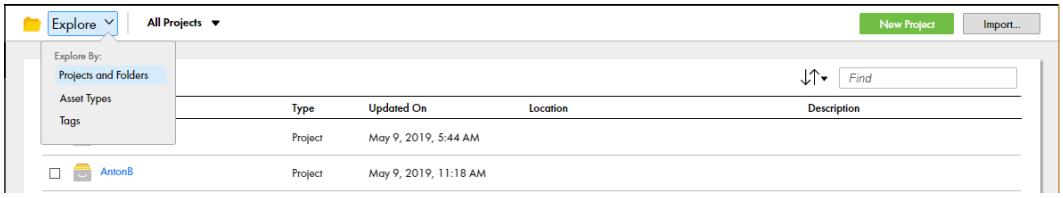
Environment Variable	Description
DBMI_WRITER_CONN_POOL_SIZE	Indicates the number of connections that a database ingestion job uses to propagate the change data to the target. The default value is 8. Valid values are 4 through 8.

Note: After you define or change an environment variable, restart the Database Ingestion Agent for the changes to take effect.

Creating projects and project folders

Create a project to contain your ingestion task assets. You can create one or more folders under the project to logically organize your assets. However, you cannot create subfolders under a folder.

1. In the Mass Ingestion service, open the **Explore** page.
2. If the **Explore** page shows objects other than projects, select **Projects and Folders** in the **Explore** menu.



3. To create a project, click **New Project**.
4. In the **New Project Properties** dialog box, enter a project name up to 255 characters in length. You can also enter an optional description of the project.

   A project name cannot contain the following characters: `#?`|{}"^&[]/\.`
5. If you want to add a folder under the project, select the project and click **New Folder**.
6. In the **New Folder Properties** dialog box, enter a folder name up to 255 characters in length. You can also enter an optional description of the project.

   A folder name cannot contain the following characters: `#?`|{}"^&[]/\.`

When you define an ingestion task, you must specify the project or project folder location to contain the task definition.

SQL Server sources

To use Microsoft SQL Server or Microsoft Azure SQL Database sources in database ingestion tasks, first prepare the source database and review the usage considerations. SQL Server sources include on-premises, RDS, and Azure SQL Managed Instance sources.

Source preparation

- For your SQL Server source type, verify that you use a SQL Server edition and version that Mass Ingestion Databases supports. See the KB article FAQ: What are the supported sources and targets for IICS Cloud Mass Ingestion service?.

- For incremental load jobs, ensure that the database user you specify in the SQL Server source connection has the db_owner role and the VIEW ANY DEFINITION privilege. To grant these privileges, use one of the following sets of SQL statements, depending on your SQL Server source type.

  For SQL Server on-premises sources, including Azure SQL Managed Instance:

  ```
  USE master;
  CREATE DATABASE <database>;
  CREATE LOGIN <login_name> WITH PASSWORD = '<password>';
  CREATE USER <user> FOR LOGIN <login_name>;
  GRANT SELECT ON master.sys.fn_dblog TO <user>;
  GRANT VIEW SERVER STATE TO <login_name>;
  GRANT VIEW ANY DEFINITION TO <login_name>;

  USE <db>;
  CREATE USER <user> FOR LOGIN <login_name>;
  EXEC sp_addrolemember 'db_owner', '<user>';
  EXEC sys.sp_cdc_enable_db
  ```

  For RDS for SQL Server:

  ```
  USE master;
  CREATE DATABASE <database>;
  CREATE LOGIN <login> WITH PASSWORD = '<password>';

  USE <database>;
  EXEC msdb.dbo.rds_cdc_enable_db '<database>';
  CREATE USER <user> FOR LOGIN <login_name>;

  USE master;
  GRANT VIEW SERVER STATE TO <login_name >;
  GRANT VIEW ANY DEFINITION TO <login_name >;

  USE <database>;
  EXEC sp_addrolemember 'db_owner', '<user>';
  ```

- For database ingestion incremental load and initial and incremental load jobs that have SQL Server sources, you must enable SQL Server Change Data Capture (CDC) on the source database.

  - For on-premises SQL Server sources, run the sys.sp_cdc_enable_db stored procedure in the database context. You must have the sysadmin role.
  - For Amazon Relational Database Service (RDS) for SQL Server sources, log in as the master user and run the msdb.dbo.rds_cdc_enable_db '*database_name*' stored procedure.

  After SQL Server CDC is enabled, SQL Server writes additional information to the transaction log and CDC tables, which Mass Ingestion Databases uses during incremental CDC processing.

  Alternatively, when you create a database ingestion task, you have the option of generating a script that enables CDC on the database and on all columns in the selected source tables. To execute the CDC script, you need to have the sysadmin role.

  Restriction: Mass Ingestion Databases cannot enable CDC for tables that contain more than 1019 columns.

Usage considerations

- Mass Ingestion Databases supports Amazon Relational Database Service (RDS) for SQL Server sources for database ingestion jobs that use any load type.
- Mass Ingestion Databases supports Azure SQL Managed Instance sources only for incremental load jobs.
- Mass Ingestion Databases does not support Microsoft Azure SQL Database sources for incremental loads and combined initial and incremental loads that use log-based CDC with the transaction log.
- Mass Ingestion Databases reads change data for incremental load jobs from the SQL Server transaction log or from the enabled SQL Server CDC tables. The change data is read from the transaction log if the required restart point (LSN) is available there. If the restart point pre-dates the active transaction log, Mass Ingestion Databases automatically transitions to reading the change data from the CDC tables instead. After reading data from the CDC tables, Mass Ingestion Databases switches back to the transaction log in a transparent

manner. The same behavior can occur if you resume an incremental load job that stopped for any reason, or if the transaction log is truncated while capture processing is down.

- When you enable CDC on the SQL Server database, SQL Server automatically creates a capture job and a cleanup job that will be executed by the SQL Server Agent. The capture job is responsible for populating the SQL Server CDC tables. The cleanup job is responsible for cleaning up records from the CDC tables. The default value for data retention in the CDC table is 72 hours, or 3 days. You can check the current retention period by running the sys.sp_cdc_help_jobs stored procedure and checking the retention value in the results. If you expect a downtime greater than 3 days, you can adjust the retention in the sys.sp_cdc_change_job stored procedure or in the SQL Server Agent cleanup job. You can also suspend the cleanup job.
- Mass Ingestion Databases supports SQL Server page compression and row compression of source data.
- Because Mass Ingestion Databases expects each source table row to be unique, Informatica recommends that each source table have a primary key. Mass Ingestion Databases does not honor unique indexes in place of a primary key. If no primary key is specified, Mass Ingestion Databases treats all columns as if they are part of the primary key.
- Mass Ingestion Databases requires read-write access on the source database. If you use SQL Server Always On availability groups, this requirement means that Mass Ingestion Databases can capture change data from the read-write primary replica but not from the read-only secondary replica.
- If a Microsoft SQL Server source uses the Always Encrypted method to encrypt column data, the CDC script that is generated from the **CDC Script** field on the **Source** page in the database ingestion task fails to run. This problem is caused by a SQL Server limitation. This problem does not occur with Transparent Data Encryption (TDE).
- Mass Ingestion Databases supports schema drift options for Microsoft SQL Server sources in database ingestion incremental load jobs. The following limitations apply:
  - Microsoft SQL Server does not support renaming tables and columns for which Change Data Capture (CDC) is enabled.
  - Microsoft SQL Server does not support changing primary keys for CDC tables.
- If source table partition changes cause rowset IDs to change, Mass Ingestion Databases can process the changes to enable database ingestion jobs to continue capturing DML changes from the tables.
- Mass Ingestion Databases does not support replicating data from SQL Server large-object (LOB) columns to Amazon Redshift in incremental and combined initial and incremental load jobs.
- Mass Ingestion Databases does not replicate data from SQL Server computed columns.
- Database ingestion incremental load and combined initial and incremental load jobs that have a SQL Server source and use a Secure Agent group with multiple agents can switch over to another agent in the group when the active agent stops running, subject to following limitations:
  - The job cannot have persistent storage enabled.
  - You must Stop and Resume the job for the switch to occur.
- If a SQL Server source is in an Always On availability group, database ingestion incremental load and combined load jobs can capture change data from transaction logs or CDC tables on the primary node or a secondary node. Also, if a node becomes unavailable, database ingestion jobs can fail over to a primary or secondary database in an availability replica to continue processing, provided that you've configured the SQL Server connection to point to an availability group listener. An availability group listener is a virtual network name (VNN) that Mass Ingestion Databases can use to access a database in an availability replica of an availability group, without having to know the SQL Server physical instance name.

Amazon Redshift targets

The following list identifies considerations for preparing and using Amazon Redshift targets:

- You can use either Amazon Redshift or Amazon Redshift Serverless targets in database ingestion jobs.
- Before writing data to Amazon Redshift target tables, database ingestion jobs stage the data in an Amazon S3 bucket. You must specify the name of the bucket when you configure the database ingestion task. The ingestion jobs use the COPY command to load the data from the Amazon S3 bucket to the Amazon Redshift target tables. For more information about the COPY command, see the Amazon Web Services documentation.
- When you define a connection for an Amazon Redshift target, provide the access key and secret access key for the Amazon S3 bucket in which you want the database ingestion jobs to stage the data before loading it to the Amazon Redshift target tables.
- Incremental load jobs and combined initial and incremental load jobs generate a recovery table named INFORMATICA_CDC_RECOVERY on the target to store internal service information. The data in the recovery table prevents jobs that are restarted after a failure from propagating previously processed data again. The recovery table is generated in the same schema as the target tables.
- Database ingestion jobs convert infinity and NaN values in FLOAT columns in source tables to null values in Amazon Redshift target tables.

Configuring connections

Configure a source connection and a target connection, if not already defined, from the **Connections** page in Administrator. Mass Ingestion Databases uses the SQL Server connector and Amazon Redshift V2 connector.

1. In Administrator, select **Connections**.
2. On the **Connections** page, click **New Connection**.
3. Configure the following connection details:

Property	Description
Connection Name	The name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Maximum length is 100 characters. Connection names are not case sensitive.
Description	A description of the connection.
Type	The type of connection, such as Amazon Redshift V2.

   After you select the connection type, additional properties that are specific to that type appear.

4. Configure the SQL Server connection properties and the Amazon Redshift connection properties.
5. To test each connection, click **Test Connection**.

   The results of the test are displayed at the top of the page.

   If a connection fails, contact the database administrator, or recheck your settings and verify that the selected runtime environment has the status of Up and Running.
6. Click **Save** to save the connection.

Microsoft SQL Server connection properties

When you set up a Microsoft SQL Server connection, configure the connection properties.

The following table describes the Microsoft SQL Server connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Type	Type of connection. Select SQL Server from the list.
Runtime Environment	The name of the runtime environment where you want to run the tasks. You cannot run a database ingestion task on a Hosted Agent or serverless runtime environment.
SQL Server Version	Microsoft SQL Server database version.
Authentication Mode	Authentication method to access Microsoft SQL Server. Select one of the following methods: - SQL Server Authentication. Uses your Microsoft SQL Server user name and password to access Microsoft SQL Server. - Windows Authentication (Deprecated). Uses the Microsoft Windows authentication to access Microsoft SQL Server. This option is available when you access Data Integration or Mass Ingestion by using Microsoft Windows. When you choose this option, you don't need to enter credentials to access Microsoft SQL Server and ensure that the user account that starts the Secure Agent service is available in the Microsoft SQL Server database. If you use Mass Ingestion Databases and want to use Windows authentication, select this option. **Note:** Windows authentication is not certified for Microsoft SQL Server 2017 version hosted on Linux. You can't configure Windows Authentication when you use a serverless runtime environment. - Active Directory Password. Uses the Azure Active Directory user name and password to authenticate and access the Microsoft Azure SQL Database. - Windows Authentication v2. Uses this authentication method to access Microsoft SQL Server from Data Integration using the agent hosted on a Linux or Windows machine. When you choose this option on Linux, enter your domain name and Microsoft Windows credentials to access Microsoft SQL Server. When you choose this option on Windows, ensure that the user account that starts the Secure Agent service is available in the Microsoft SQL Server database. - Kerberos. Uses Kerberos authentication to connect to Microsoft SQL Server. When you choose this option on Windows, ensure that the user account that starts the Secure Agent service is available in the Microsoft SQL Server database. You don't need to enter your credentials to access Microsoft SQL Server.
Domain	Applies to Windows Authentication v2. The domain name of the Windows user.

Property	Description
User Name	User name for the database login. The user name can't contain a semicolon. To connect to Microsoft Azure SQL Database, specify the user name in the following format: `username@host` If you use Windows Authentication v2, the Microsoft SQL server driver ignores the user name specified in this field and use the Windows NT user name. **Note:** This property is not applicable if you use the Windows Authentication mode to access Microsoft SQL Server.
Password	Password for the database login. The password can't contain a semicolon. If you use Windows Authentication v2, the Microsoft SQL server driver ignores the password specified in this field and use the Windows NT password. **Note:** This property is not applicable if you use the Windows Authentication mode to access Microsoft SQL Server.
Host	Name of the machine hosting the database server. To connect to Microsoft Azure SQL Database, specify the fully qualified host name. For example, `vmjcmwxsfboheng.westus.cloudapp.azure.com`.
Port	Network port number used to connect to the database server. Default is 1433.
Instance Name	Instance name of the Microsoft SQL Server database.
Database Name	Database name for the Microsoft SQL Server target connection. Database name is case-sensitive if the database is case-sensitive. Maximum length is 100 characters. Database names can include alphanumeric and underscore characters.
Schema	Schema used for the target connection.
Code Page	The code page of the database server.
Encryption Method	The method that the Secure Agent uses to encrypt the data sent between the driver and the database server. You can use the encryption method to connect to Microsoft Azure SQL Database. Default is None.
Crypto Protocol Version	Cryptographic protocols to use when you enable SSL encryption.
Validate Server Certificate	When set to True, Secure Agent validates the certificate that is sent by the database server. If you specify the HostNameInCertificate parameter, Secure Agent also validates the host name in the certificate. When set to false, the Secure Agent doesn't validate the certificate that is sent by the database server.
Trust Store	The location and name of the truststore file. The truststore file contains a list of Certificate Authorities (CAs) that the driver uses for SSL server authentication.
Trust Store Password	The password to access the contents of the truststore file.
Host Name in Certificate	Host name of the machine that hosts the secure database. If you specify a host name, the Secure Agent validates the host name included in the connection with the host name in the SSL certificate.

Property	Description
Metadata Advanced Connection Properties	Additional properties for the JDBC driver to fetch the metadata. If you specify more than one property, separate each key-value pair with a semicolon.
Runtime Advanced Connection Properties	Additional properties for the ODBC driver to run mappings. If you specify more than one property, separate each key-value pair with a semicolon.

Amazon Redshift V2 connection properties

When you set up an Amazon Redshift V2 connection, configure the connection properties.

The following table describes the Amazon Redshift V2 connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Type	The Amazon Redshift V2 connection type.
Runtime Environment	Name of the runtime environment where you want to run the tasks. You cannot run an application ingestion task, database ingestion task, file ingestion task, or streaming ingestion task on a Hosted Agent or serverless runtime environment.
Authentication Type	The authentication method that the connector must use to log in to Amazon Redshift. Select one of the following authentication types: - Default. Uses the username and password to connect to Amazon Redshift. - Redshift IAM Authentication via AssumeRole. Uses AssumeRole for IAM authentication to connect to Amazon Redshift. Doesn't apply to Mass Ingestion application and database ingestion tasks.
Username	User name of the Amazon Redshift account.
Password	Password for the Amazon Redshift account. Applies only to default authentication.
Use EC2 Role to Assume Role	Optional. Select the check box to enable the EC2 role to assume another IAM role specified in the IAM Role ARN option. **Note:** The EC2 role must have a policy attached with a permission to assume an IAM role from the same or different account. Does not apply to application ingestion and database ingestion tasks. By default, this check box is not selected.

Property	Description
S3 Access Key ID	Access key to access the Amazon S3 staging bucket. Enter the access key value based on the following authentication methods: - Basic authentication. Enter the actual access key value. - IAM authentication. Do not enter the access key value. - Temporary security credentials using assume role. Enter access key of an IAM user with no permissions to access the Amazon S3 staging bucket. - Assume role for EC2. Do not enter the access key value. **Note:** If you use the connection for application ingestion or database ingestion tasks that use key-based authentication, provide the access key value.
S3 Secret Access Key	Secret access key to access the Amazon S3 staging bucket. The secret key is associated with the access key and uniquely identifies the account. Enter the access key value based on the following authentication methods: - Basic authentication. Enter the actual access secret value. - IAM authentication. Do not enter the access secret value. - Temporary security credentials using assume role. Enter access secret of an IAM user with no permissions to access Amazon S3 staging bucket. - Assume role for EC2. Do not enter the access secret value. **Note:** If you use the connection for application ingestion or database ingestion tasks that use key-based authentication, provide the access key value.
S3 IAM Role ARN	The Amazon Resource Number (ARN) of the IAM role assumed by the user to use the dynamically generated temporary security credentials. Set the value of this property if you want to use the temporary security credentials to access the Amazon S3 staging bucket. For more information about how to get the ARN of the IAM role, see the AWS documentation. **Note:** If you use the connection for application ingestion or database ingestion tasks that uses role-based authentication, but not the default role for the AWS cluster, specify an IAM role ARN. If you use the default role, leave this field blank.
External Id	The external ID for a more secure access to the Amazon S3 bucket when the Amazon S3 staging bucket is in a different AWS account. Does not apply to application ingestion and database ingestion tasks.
Master Symmetric Key	A 256-bit AES encryption key in the Base64 format when you enable client-side encryption. You can generate a key using a third-party tool. Does not apply to application ingestion and database ingestion tasks.
JDBC URL	The URL of the Amazon Redshift V2 connection. Enter the JDBC URL in the following format: `jdbc:redshift://<amazon_redshift_host>:<port_number>/<database_name>`

Property	Description
Cluster Region	The AWS cluster region in which the bucket you want to access resides. Select a cluster region if you choose to provide a custom JDBC URL that does not contain a cluster region name in the **JDBC URL** connection property. If you select a cluster region in both **Cluster Region** and **JDBC URL** connection properties, the agent ignores the cluster region that you specify in the **JDBC URL** connection property. To use the cluster region name that you specify in the **JDBC URL** connection property, select **None** as the cluster region in this property. You can only read data from or write data to the cluster regions supported by AWS SDK. Select one of the following cluster regions: - None - Asia Pacific(Mumbai) - Asia Pacific(Seoul) - Asia Pacific(Singapore) - Asia Pacific(Sydney) - Asia Pacific(Tokyo) - Asia Pacific(Hong Kong) - AWS GovCloud (US) - AWS GovCloud (US-East) - Canada(Central) - China(Bejing) - China(Ningxia) - EU(Ireland) - EU(Frankfurt) - EU(Paris) - EU(Stockholm) - South America(Sao Paulo) - Middle East(Bahrain) - US East(N. Virginia) - US East(Ohio) - US West(N. California) - US West(Oregon) Default is **None**. Required for Mass Ingestion application and database ingestion tasks.
Customer Master Key ID	The customer master key ID generated by AWS Key Management Service (AWS KMS) or the ARN of your custom key for cross-account access. You must generate the customer master key ID for the same region where your Amazon S3 staging bucket resides. You can either enter the customer-generated customer master key ID or the default customer master key ID. You can use a cross account KMS key for the same regions when you run mappings in advanced mode. Doesn't apply to application ingestion and database ingestion tasks.

Amazon Redshift AssumeRole authentication

Configure the following properties when you select the Redshift AssumeRole authentication type:

Property	Description
Cluster Identifier	The unique identifier of the cluster that hosts Amazon Redshift for which you are requesting the security credentials. Specify the cluster name.
Database Name	Name of the Amazon Redshift database.

Property	Description
Database Group	Name of the database group to which an IAM user will be added. You can add multiple database groups separated by a comma. **Note:** If you do not specify a group, the user is added to the public group.
Expiration Time	The time duration that the password for the Amazon Redshift database user expires. Specify a value between 900 seconds and 3600 seconds. Default is 900.
Auto Create DBUser	Select to create a new Amazon Redshift database user at run time. Default is disabled.
Redshift IAM Role ARN	The Amazon Resource Number (ARN) of the IAM role assumed by the user or EC2 to use the dynamically generated temporary security credentials. Set the value of this property if you want to use the temporary security credentials to access Amazon Redshift. For more information about how to get the ARN of the IAM role, see the AWS documentation.
Redshift Access Key ID	The access key of the IAM user that has permissions to assume the Redshift IAM AssumeRole ARN.
Redshift Secret Access Key	The secret access key of the IAM user that has permissions to assume the Redshift IAM Assume Role ARN.

Note: In an Amazon Redshift V2 connection where you write to a target, and the target connection has **Auto Create DBuser** enabled, the new user cannot truncate a table created by an existing user.

Configuring a database ingestion task

In Mass Ingestion, use the database ingestion task wizard to configure a database ingestion task.

On the wizard pages, complete the following configuration tasks:

1. Define basic task information, such as the task name, project location, runtime environment, and load type.
2. Configure the source.
3. Configure the target.
4. Configure runtime options.

Click **Next** or **Back** to navigate from one page to another. At any point, you can click **Save** to save the information that you have entered so far.

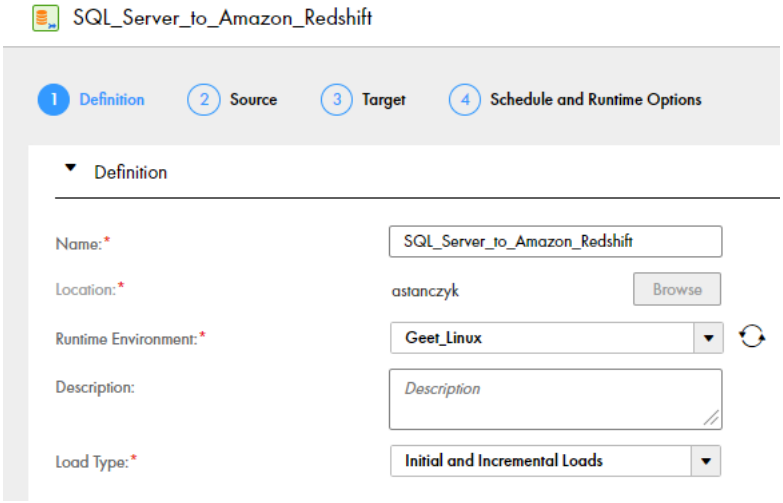
After you complete all wizard pages, save the information and then click **Deploy** to make the task available as an executable job to the Secure Agent.

Step 1: Defining basic task information

To begin defining a database ingestion task, you must first enter some basic information about the task, such as a task name, project or project folder location, and load operation type.

1. In Mass Ingestion, click **New** > **Database Ingestion Task**.

   The **Definition** page of the Mass Ingestion Databases Task wizard appears.



2. Configure the following properties:

Property	Description
Name	A name for the database ingestion task. Task names can contain Latin alphanumeric characters, spaces, periods (.), commas (,), underscores (_), plus signs (+), and hyphens (-). Task names cannot include other special characters. Task names are not case sensitive. Maximum length is 50 characters. **Note:** If you include spaces in the database ingestion task name, after you deploy the task, the spaces do not appear in the corresponding job name.
Location	The project or project\folder that will contain the task definition. The default is the currently selected project or project subfolder in Explore. If a project or project subfolder is not selected, the default is the **Default** project.
Runtime Environment	The runtime environment in which you want to run the task. The runtime environment must be a Secure Agent group that consists of one or more Secure Agents. A Secure Agent is a lightweight program that runs tasks and enables secure communication. You cannot use the Hosted Agent or a serverless runtime environment. **Tip:** Click the **Refresh** icon to refresh the list of runtime environments.

Property	Description
Description	An optional description for the task. Maximum length is 4,000 characters.
Load Type	The type of load operation that the database ingestion task performs. For this scenario, select **Initial and Incremental Loads** to perform an initial load of point-in-time data to the target and then automatically switch to propagating incremental data changes made to the same source tables on a continuous basis. **Note:** If a change record is captured during the initial unload load phase, it's withheld from apply processing until after the unload phase completes. Any insert rows captured during the unload phase are converted into a pair of delete and insert operations so that only one insert row is applied to the target in the case where the insert occurs in both the unloaded data and the captured change data.

3. Click **Next**.

Step 2: Configuring the SQL Server source

Configure the SQL Server source on the **Source** page of the database ingestion task wizard.

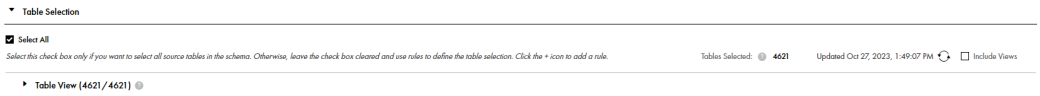
1. In the **Connection** list, select the connection for the SQL Server source that you configured and tested in Administrator for a runtime environment that your organization uses. The connection type appears in parentheses after the connection name.

   The list includes only the connection types that are valid for the load type selected on the **Definition** page. No connections are listed if you did not select a load type.

   Note: After you deploy the ingestion task, you cannot change the connection without first undeploying the associated ingestion job. After you change the connection, you must deploy the task again.
2. In the **Schema** list, select the source schema that includes the source tables.

   When creating a task that has a SQL Server source, the schema name that is specified in the connection properties is displayed by default.
3. Under **Change Data Capture Method**, set the **CDC Method** field to **Log-based CDC**.

   In this CDC method, Inserts, Updates, Deletes, and column DDL changes are captured from the SQL Server transaction log and the enabled SQL Server CDC tables.
4. Under **Table Selection**, select **Select All** to select all tables and columns in the schema for data replication. The **Tables Selected** field shows the count of all selected tables. You won't be able to edit the selection later under **Table View**.



   Note: Fetching information for all of the tables might take a long time.

   Alternatively, you can use rules to define a subset of source tables to replicate.
5. To use rules to select the source tables, make sure that the **Select All** check box is cleared and then add rules.

   When rule-based selection is used, you can refine the set of selected tables by table under **Table View** and also set an option for trimming spaces in character data.

Note: The default "Include *" rule selects all tables in the specified schema. To see how many tables are selected by this rule, click the Refresh icon to display the table count in **Total Tables Selected** and click **Apply Rules** to see the table count in **Table View**.

To add a rule:

a. Under **Rules**, click the Add Rule (+) icon above the first table. A row is added to the table.

b. In the **Table Rule** column, select **Include** or **Exclude** to create an inclusion or exclusion rule, respectively.

c. In the **Condition** column, enter a table name or a table-name mask that includes one or more wildcards to identify the source tables to include in or exclude from table selection. Use the following guidelines:

- A mask can contain one or both of the following wildcards: an asterisk (*) wildcard to represent one or more characters and a question mark (?) wildcard to represent a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
- The task wizard is case sensitive. Enter the table names or masks in the case with which the tables were defined.
- Do not include delimiters such as quotation marks or brackets, even if the source database uses them.
- If a table name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

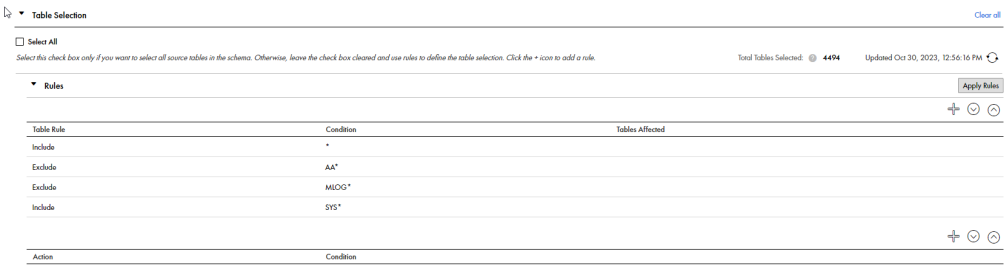
d. Define additional rules as needed.

The rules will be processed in the order in which they're listed, from top to bottom. Use the arrow icons to change the order.

e. When finished, click **Apply Rules**.

The **Total Tables Selected** and **Tables Affected** counts are updated.

The following image shows multiple rules defined on the **Source** page:



After you apply rules, if you add, delete, or change rules, you must click **Apply Rules** again. Click the Refresh icon to update the table counts. If you delete all rules without clicking **Apply Rules**, a validation error will occur at deployment, even if the **Table View** list still lists tables. If you switch to **Select All**, the rules are no longer in effect and disappear.

6. To perform trim actions on character columns in the source tables selected based on rules, create column action rules.

To create a column action rule:

a. Click the Add Rule (+) icon above the second Action table under **Rules**.

b. In the **Action** column, select one of the following options:

- **LTRIM**. Trims spaces to the left of character column values.
- **RTRIM**. Trims spaces to the right of character column values.
- **TRIM**. Trims spaces to the left of and to the right of character column values.

c. In the **Condition** column, enter a column name or a column name mask that includes one or more asterisk (*) or question mark (?) wildcards. The value is matched against columns in the selected source tables to identify the columns to which the action applies.

Note: You can define multiple rules for different action types or for the same action type with different conditions. The rules are processed in the order in which they're listed, from top to bottom. Use the arrow icons to change the order.

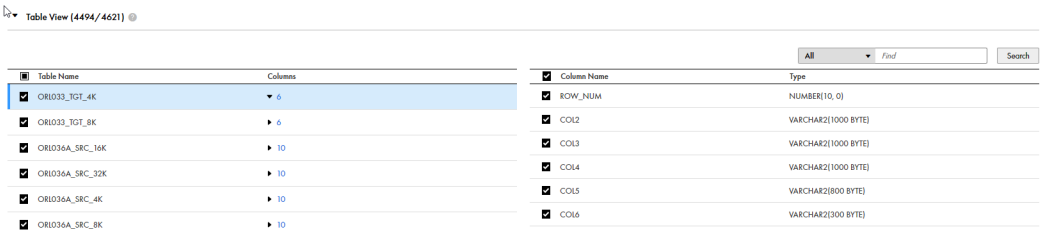
7. Under **Table View**, view or edit the set of selected source tables and columns.

   If you selected **Select All**, the lists of tables and columns are view only.

   If you applied rules, you can refine the set of selected tables by clicking the check box next to individual tables. Deselect any tables that you do not want to replicate, or select additional items to replicate. Click the Refresh icon to update the selected tables count.

   You can also individually deselect or reselect the columns in a selected source table. To view or change the columns from which data will be replicated for a selected table, click the highlighted number of columns in the **Columns** column. The column names and data types are displayed to the right. By default, all columns are selected for a selected source table. To deselect a column or reselect it, click the check box next to the column name. You cannot deselect a primary key column.

   The following image shows selected tables and the selected columns for the first table:



Notes:

- To search for tables or columns or both, in the drop-down list above the columns list, select **Table Name**, **Columns**, or **All** and then enter a search string in the *Find* box and click **Search**. You can include a single asterisk (*) wildcard at the beginning or end of the string.
- The first time you change a check box setting for a table or column, the rules are no longer in effect. The selections under **Table View** take precedence. However, if you click the Add Rule (+) icon again, any tables that you deselected or selected individually are reflected as new rules in the Rules list and the rules once again take precedence. If you want to return to the **Table View** list, click **Apply Rules** again.
- In the **Table View** section, the length of character columns is shown in bytes. Since the number of bytes per character differs depending on the character set encoding used by the database, for some sources, the actual column length in number of characters might be different than the number of bytes shown in the **Table View**.
- If you select columns individually, the resulting set of columns is fixed and will not be updated by any schema change, regardless of schema drift settings. For example, if a source column is added or renamed, that column is silently excluded from CDC processing because it's not in the list of selected columns. However, if a selected column is dropped on the source, the schema drift Drop Column option controls how it's handled. The dropped column operation is not reflected in the list of columns until you apply rules again.

8. If one or more of the selected source tables are not enabled for change data capture, you can generate a script for enabling CDC and then run or download the script.
    a. In the **CDC Script** field, select **Enable CDC for all columns** to enable CDC for all columns in the selected source tables. This option is the only valid option for SQL Server sources. Do not use the **Enable CDC for primary key columns** option for SQL Server sources.

       The script runs the sys.sp_cdc_enable_db and sys.sp_cdc_enable_table stored procedures to enable CDC on the source database and tables. For RDS for SQL Server, the script runs the msdb.dbo.rds_cdc_enable_db procedure to enable CDC on the source database, and the sys.sp_cdc_enable_table script to track CDC for tables.
    b. To run the script, click **Execute**.

       If you do not have a database role or privilege that allows you to run the script, click the Download icon to download the script. The script file name has the following format: `cdc_script_taskname_number.txt`. Then ask your database administrator to run the script.

       Make sure the script runs before you run the database ingestion task.

    Note: If you change to the **CDC Script** option later and run the script again, the script first drops CDC for the original set of columns and then enables CDC for the current set of columns.
9. In the **Capture Filegroup** field, enter the name of the filegroup to be used for the change table that is created for the capture. If you leave this field empty, the change table is located in the default filegroup of the database.
10. In the **Gating Role** field, enter the name of the database role that is used to gate access to change data. If you leave this field empty, the database does not use the gating role.
11. To create and download a list of the source tables that match the table selection criteria, perform the following substeps:
    a. If you used rule-based table selection, in the **List Tables by Rule Type** list, select the type of selection rules that you want to use. Options are:
       - **Include Rules Only**
       - **Exclude Rules Only**
       - **Include And Exclude Rules**
    b. To include columns in the list, regardless of which table selection method you used, select the **Include Columns** check box.
    c. Click the Download icon.

       A downloaded list that includes columns has the following format:

       ```
       status,schema_name,table_name,object_type,column_name,comment
       ```

The following table describes the information that is displayed in the downloaded list:

Field	Description
status	Indicates whether Mass Ingestion Databases excludes the source table or column from processing because it has an unsupported type. Valid values are: - **E**. The object is excluded from processing by an Exclude rule. - **I**. The object is included in processing. - **X**. The object is excluded from processing because it is an unsupported type of object. For example, unsupported types of objects include columns with unsupported data types and tables that include only unsupported columns. The comment field provides detail on unsupported types.
schema_name	Specifies the name of the source schema.
table_name	Specifies the name of the source table.
object_type	Specifies the type of the source object. Valid values are: - **C**. Column. - **T**. Table.
column_name	Specifies the name of the source column. This information appears only if you selected the **Columns** check box.
comment	Specifies the reason why a source object of an unsupported type is excluded from processing even though it matches the selection rules.

12. Under **Advanced**, set the advanced properties that are available for combined initial and incremental load jobs with SQL Server sources.

Field	Description
Enable Persistent Storage	Select this check box to enable persistent storage of transaction data in a disk buffer so that the data can be consumed continually, even when the writing of data to the target is slow or delayed. Benefits of using persistent storage are faster consumption of the source transaction logs, less reliance on log archives or backups, and the ability to still access the data persisted in disk storage after restarting a database ingestion job.
Enable Partitioning	Select this check box to enable partitioning of source objects. When an object is partitioned, the database ingestion job processes the records read from each partition in parallel. For SQL Server sources, partitioning is based on the primary key. **Note:** In combined initial and incremental loads, the partitioning of source objects occurs only in the initial load phase.

Field	Description
Number of Partitions	If you enable partitioning of source objects, enter the number of partitions you want to create. The default number is 5. The minimum value is 2.
Initial Start Point for Incremental Load	Set this field if you want to customize the position in the source logs from which the database ingestion job starts reading change records the first time it runs. Options are: - **Earliest Available**. The earliest available record in the active transaction log. - **Latest Available**. The latest available position in the database log. - **Position**. A valid SQL Server LSN that is used to determine a position in the change stream from which to start retrieving change records. The LSN must be equal to or less than the current LSN. An invalid value will cause the job to fail. The default value is 0, which causes processing to start from the earliest available position. A non-zero LSN that predates the beginning of the active transaction log causes data to be read from the CDC tables instead of from the transaction log. - **Specific Date and Time**. A date and time, in the format MM/DD/YYYY hh:mm AM|PM, that Mass Ingestion Databases uses to determine the position in the change stream from which to start retrieving change records. Mass Ingestion Databases retrieves only the changes that were started after this date and time. If you enter a date and time earlier than the earliest date and time in the available archived logs, the job will fail. The default is **Latest Available**. **Notes:** - The **Initial Start Point for Incremental Load** option pertains only to the initial run of a job. Thereafter, if you resume a stopped or aborted job, the job begins propagating source data from where it last left off. - For combined initial and incremental load jobs, initial loading is not performed until the incremental processing of change data reaches the end of the current transaction log. For this reason, Informatica recommends that you select **Latest Available** as the start point. For SQL Server, data changes are read from the active portion of the transaction log if the requested LSN is available there. If the LSN predates the active transaction log, the data changes are read from previously enabled CDC tables. Ensure that SQL Server CDC is enabled on the source tables.

13. Under **Custom Properties**, you can specify custom properties at the direction of Informatica Global Customer Support to meet your special requirements. In this scenario, skip this step.
14. Click **Next**.

Step 3: Configuring the Amazon Redshift target

Configure the target on the **Target** page of the database ingestion task wizard.

1. In the **Connection** list, select the Amazon Redshift connection that you previously defined and tested in Administrator for the runtime environment. The connection type appears in parentheses after the connection name.

   The list includes only the connection types that are valid for the load type selected on the **Definition** page. No connections are listed if you did not select a load type.

   If you change the load type and the selected connection is no longer valid, a warning message is issued and the **Connection** field is cleared. You must select another connection that is valid for the updated load type.

   Note: After you deploy the ingestion task, you cannot change the connection without first undeploying the associated ingestion job. You must then deploy the task again.
2. In the **Target** section, configure the properties that pertain to an Amazon Redshift target.

The following table describes these properties:

Property	Description
Target Creation	The only available option is **Create Target Tables**, which generates the target tables based on the source tables. **Note:** After the target table is created, Mass Ingestion Databases intelligently handles the target tables on subsequent job runs. Mass Ingestion Databases might truncate or re-create the target tables depending on the specific circumstances.
Schema	Select the target schema in which Mass Ingestion Databases creates the target tables.
Bucket	Specifies the name of the Amazon S3 bucket that stores, organizes, and controls access to the data objects that you load to Amazon Redshift.
Directory	Specifies the subdirectory where Mass Ingestion Databases stores output files for this job.

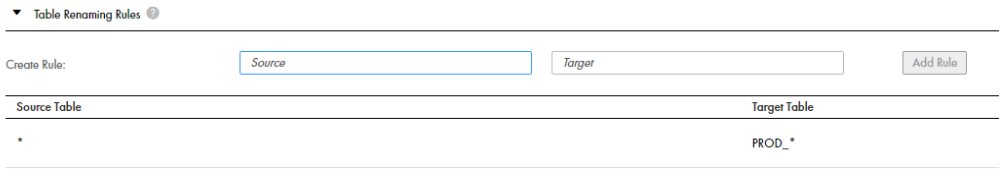
3. If you want to rename the target objects that are associated with the selected source tables, define table renaming rules. For example, you might want to add a prefix or suffix. To create a rule for renaming tables:
   a. Under **Table Renaming Rules**, in the **Create Rule** fields, enter a source table name or a table name mask that includes one or more wildcards. Then enter the corresponding target table name or table name mask.

      Notes:

      - For the source, you can enter only the asterisk (*) wildcard to select all source tables that match the table selection criteria on the **Source** page. Or you can enter a specific source table name or a table-name mask that includes one or more of the following wildcards: an asterisk (*) to represent one or more characters or a question mark (?) to represent a single character.
      - To use a table-name mask with the wildcard character for the target, you must also use the wildcard character in the source. If you use a specific source table name with a target table mask that includes the wildcard character, the task deployment will fail.
      - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).

   b. Click **Add Rule**

      The rule appears in the rules list.



   You can define multiple table rules. The order of the rules does not matter with regard to how they are processed unless a table matches multiple rules. In this case, the last matching rule determines the name of the table.

4. Under **Custom Properties**, you can specify custom properties at the direction of Informatica Global Customer Support to meet your special requirements. In this scenario, skip this step.
5. Click **Next** if available, or click **Save**.

Step 4: Configuring runtime options

On the **Schedule and Runtime Options** page of the database ingestion task wizard, you can configure some runtime options.

1. Under **Schema Drift Options**, specify the schema drift option to use for each of the supported types of DDL operations.

   The types of supported DDL operations are:

   - Add Column
   - Modify Column
   - Drop Column
   - Rename Column

   The following table describes the schema drift options that you can set for a DDL operation type:

Option	Description
Ignore	Do not replicate DDL changes that occur on the source database to the target. For Amazon Redshift targets, this option is the default option for the Drop Column and Rename Column operation types.
Replicate	Replicate the DDL operation to the target. For Amazon Redshift targets, this option is the default option for the Add Column and Modify Column operation types. **Restrictions:** - If you try to replicate a type of schema change that is not supported on the target, database ingestion jobs associated with the task will end with an error. - Add Column operations that add a primary-key column are not supported and can cause unpredictable results. - Modify Column operations that change the NULL or NOT NULL constraint for a column are not replicated to the target by design because changing the nullibility of a target column can cause problems when subsequent changes are applied. - DDL operations are replicated to the target only after some data changes have been captured.
Stop Job	Stop the entire database ingestion job.
Stop Table	Stop processing the source table on which the DDL change occurred. When one or more of the tables are excluded from replication because of the **Stop Table** schema drift option, the job state changes to **Running with Warning**. **Important:** The database ingestion job cannot retrieve the data changes that occurred on the source table after the job stopped processing it. Consequently, data loss might occur on the target. To avoid data loss, you will need to resynchronize the source and target objects that the job stopped processing. Use the **Resume With Options** > **Resync** option.

2. Under **Advanced**, optionally edit the **Number of Rows in Output File** value to specify the maximum number of rows that the database ingestion task writes to an output data file for the Amazon Redshift target.

   Change data is flushed to the target either when this number of rows is reached or when the flush latency period expires and the job is *not* in the middle of processing a transaction. The flush latency period is the time that the job waits for more change data before flushing data to the target. The latency period is internally set to 10 seconds and cannot be changed.

   Valid values are 1 through 100000000. The default value is 100000 rows.

3. Under **Schedule**, the scheduling options are unavailable for combined initial and incremental load tasks. Skip this field.

4. Under **Custom Properties**, you can specify custom properties that Informatica provides to meet your special requirements. Specify these properties only at the direction of Informatica Global Customer Support.

5. Click **Save**.

Deploying the database ingestion task

After you define a database ingestion task and save it, deploy the task to create an executable job instance on the on-premises system that contains the Secure Agent and the Database Ingestion agent service and DBMI packages. You must deploy the task before you can run the job. The deploy process also validates the task definition.

1. In the database ingestion task wizard, save the completed task definition.
2. To deploy a task, click **Deploy**.

After you deploy a task successfully, the associated job instance has the status of Deployed. You can run the job from either the **My Jobs** page in Mass Ingestion or from the **All Jobs** tab on the **Mass Ingestion** page in Operational Insights.

Deployment considerations:

- If you included spaces in the database ingestion task name, the spaces are omitted from the corresponding job name for the generated job instance.
- If the deployment process fails, the job status switches to Failed. You can then Undeploy the job. To diagnose the error, download the error log from the **My Jobs** page in Mass Ingestion or from the **All Jobs** tab on the **Mass Ingestion** page in Operational Insights. In the Actions menu for the job, click **Error Log**. After you resolve the problem, deploy the task again from the database ingestion task wizard.
- If you undeploy a job and then want to run a job for the associated ingestion task again, you must deploy the task again to create a new job instance. The new job instance name ends with an incremented number in the format *taskname-job_instance_number*. The job instance number is incremented each time you deploy the ingestion task by adding 1 to the maximum instance number across all ingestion jobs.
- If the Secure Agent is restarted while the task is deploying, the job status switches to Failed. Avoid restarting the Secure Agent while tasks are being deployed.
- If a task appears to be hung in the Deploying state, restart the Secure Agent. The associated job instance acquires the status of Failed. You can undeploy it and then deploy it again.

Running the deployed database ingestion job

You can run a database ingestion job after it has been deployed and is in a state other than Undeployed from either the **My Jobs** page in the Mass Ingestion service or from the **All Jobs** tab on the **Mass Ingestion** page in Operational Insights.

1. Navigate to the row for the job that you want to run.
2. In the **Actions** menu at the right end of the row, click **Run**.

   A subtask is started for each source table.

Monitoring database ingestion jobs

You can monitor the progress, performance, and status of ingestion jobs from the Mass Ingestion and Operational Insights services.

Depending on the service you use and type of ingestion job, you can view following monitoring information:

- On the **My Jobs** page in the Mass Ingestion service, monitor the ingestion jobs for the ingestion tasks that you deployed. You can view a list of your jobs that includes general job properties such as the task type, runtime environment, start time, duration, and current job status.

- On the **Mass Ingestion** page in Operational Insights service, monitor *all* types of ingestion jobs that any member of your organization deployed. You can view the following types of information:
  - Summary counts of ingestion jobs by task type and job status.
  - Recent jobs that require your attention because they failed or are running with errors or warnings.
  - A list of all ingestion jobs by type, including the general job properties.
- From either the list of your jobs or the list of all jobs, you can drill down to details for a specific job by clicking the job name. You can view overview job information, source object processing details, and alerts.

Note: Usually, a job name corresponds to the ingestion task name. For database ingestion jobs, the job name has the format *taskname-job_instance_number*. The number is incremented each time the job is deployed.

Other resources

For more information, visit the Informatica Documentation Portal .

You can also access the latest Mass Ingestion and Administrator help information from within the Informatica Intelligent Cloud Services interface.

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